# Proving Basic Algorithms Correct: Sorting

Computer-Assisted Theorem Proving (7CCSACTP)

Mohammad Abdulaziz

# Principle: Count function calls

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# Principle: Count function calls

For every function $f :: \tau_1 \Rightarrow \cdots \Rightarrow \tau_n \Rightarrow \tau$
define a *timing function* $t_f :: \tau_1 \Rightarrow \cdots \Rightarrow \tau_n \Rightarrow \text{nat}$:

Translation of defining equations:

$$\frac{e \rightsquigarrow e'}{f\, p_1 \ldots p_n = e \;\rightsquigarrow\; t_f\, p_1 \ldots p_n = e' + 1}$$

Translation of expressions:

$$\frac{s_1 \rightsquigarrow t_1 \quad \ldots \quad s_k \rightsquigarrow t_k}{g\, s_1 \ldots s_k \rightsquigarrow t_1 + \cdots + t_k + t_g\, s_1 \ldots s_k}$$

All other operations (variable access, constants, constructors, primitive operations on `bool` and numbers) cost 1

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# Example

```
app [] ys = ys
```

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$\rightsquigarrow$

$$\texttt{t_app [] ys} = \underbrace{1}_{\texttt{ys}} + \underbrace{1}_{\text{call of } \texttt{app}}$$

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app (x#xs) ys = x # app xs ys
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app (x#xs) ys = x # app xs ys
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$\rightsquigarrow$

$$\texttt{t_app (x\#xs) ys} = \underbrace{1}_{\texttt{x}} + (\underbrace{1}_{\texttt{xs}} + \underbrace{1}_{\texttt{ys}} + \underbrace{\texttt{t_app xs ys}}_{\text{recursive call}}) + \underbrace{1}_{\texttt{\#}} + \underbrace{1}_{\text{call of } \texttt{app}}$$

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# A compact formulation of $e \rightsquigarrow t$

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$t$ is the sum of all $t_g\ s_1 \ldots s_k$
such that $g\ s_1 \ldots s_k$ is a subterm of $e$

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# A compact formulation of $e \rightsquigarrow t$

$t$ is the sum of all $t_g\ s_1 \ldots s_k$
such that $g\ s_1 \ldots s_k$ is a subterm of $e$

If $g$ is

- a variable, a constant, a constructor or
- a predefined function on `bool` or numbers

then $t_g \cdots = 1$.

# *if* and *case*

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So far we model a call-by-value semantics

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Conditionals and case expressions are evaluated lazily.

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Translation:

$$\frac{b \rightsquigarrow t \quad s_1 \rightsquigarrow t_1 \quad s_2 \rightsquigarrow t_2}{\textbf{if } b \textbf{ then } s_1 \textbf{ else } s_2 \rightsquigarrow t + (\textbf{if } b \textbf{ then } t_1 \textbf{ else } t_2)}$$

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Similarly for **case**

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t_app (x#xs) ys = t_app xs ys + 1
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$\Longrightarrow$ Count only

- the defined functions via $t_f$ and
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## Example

```
t_app (x#xs) ys = t_app xs ys + 1
```

$\Longrightarrow$ Count only

- the defined functions via $t_f$ and
- 1 for the function call.

All other operations (variables etc) cost 0, not 1.

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# Discussion

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- The definition of $t_f$ from $f$ can be automated.

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- The correctness of $t_f$ could be proved w.r.t. a semantics that counts computation steps.

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# Discussion

- The definition of $t_f$ from $f$ can be automated.
- The correctness of $t_f$ could be proved w.r.t.
  a semantics that counts computation steps.
- Precise complexity bounds (as opposed to $O(\cdot)$) would require a formal model of (at least) the compiler and the hardware.

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